



Artificial Intelligence—Derived Biomechanical Index for Early Corneal Ectasia Detection: Advancing Beyond Tomography

Hazem Abdelmotaal, MD,¹ Suphi Taneri, MD,^{2,3} Ramin Salouti, MD,⁴ M. Hossein Nowroozzadeh, MD,⁴ Ali H. Al-Timemy, PhD,⁵ Alexandru Lavric, PhD,⁶ Mostafa El Habib Daho, PhD,^{7,8} Hidenori Takahashi, MD, PhD,^{9,10} Rossen Mihaylov Hazarbasanov, MD,¹¹ Siamak Yousefi, PhD^{12,13}

Purpose: To develop and evaluate a novel artificial intelligence (AI)—derived metric for detection of early ectasia (E) by leveraging spatiotemporal biomechanical data derived from dynamic cornea videos only.

Design: Multicenter cross-sectional case-control retrospective study.

Participants: A total of 451 eyes from 451 patients from 2 centers with both Scheimpflug tomography and dynamic corneal deformation examinations.

Methods: The training data set included 167 normal (N) patients with stable eyes postlaser vision correction, 83 NT eyes with VAE (VAE-NT), and better eyes from 64 patients with bilateral keratoconus (ie, mild KC [MKC]). The external validation data set included 68, 33, and 36 patients from the N, VAE-NT, and MKC groups, respectively. Extensive spatial and temporal pixel-level analysis of corneal displacement, stromal gray scale changes, and thickness variations from Corvis ST video frames was performed to identify a novel risk score with best detection of early E (combined VAE-NT and MKC). Findings were further validated using various approaches based on the external data set.

Main Outcome Measures: Area under the receiver operating characteristics curve (AUC), accuracy, specificity, and sensitivity for detecting early E.

Results: The novel AI-derived index achieved an AUC of 0.995 with accuracy, sensitivity, and specificity of 96%, 97%, and 96%, respectively, outperforming all Scheimpflug tomographic and dynamic deformation combined indices such as Belin/Ambrósio Enhanced Ectasia Display, Corvis Biomechanical Index (CBI), and Tomographic and Biomechanical Index (TBI).

Conclusions: The AI-derived index based on spatiotemporal corneal biomechanical data outperformed existing instrument's topographic, tomographic, and biomechanical indices, in detecting E. This novel metric offers a clinically meaningful enhancement in early diagnosis and management of corneal E disease.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science* 2026;6:100952 © 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).



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Keratoconus (KC), the most common ectatic corneal disease, is characterized by progressive corneal thinning and steepening, leading to irregular astigmatism, myopia, and a decline in vision-related quality of life.¹ Early diagnosis is critical to prevent disease progression and minimize the risk of postrefractive surgery ectasia (E).² Although corneal topography and tomography are essential tools for detecting subtle corneal shape abnormalities, they may not detect the earliest biomechanical changes.³ Reports of post laser vision correction (LVC) E in patients with seemingly normal topography (NT) and tomography underscore this limitation.⁴

Studies suggest that ectatic corneas are biomechanically weaker than healthy corneas, with focal biomechanical changes often preceding detectable topographic and

tomographic changes.^{3,5} Notably, biomechanical abnormalities have been observed in both eyes of patients with unilateral KC, a condition termed “very asymmetric ectasia” (VAE).⁶ Consequently, in vivo corneal biomechanical assessment has emerged as a promising approach for detecting early E, particularly in topographically normal (N) eyes.^{7,8} Analyzing the seemingly unaffected eye in VAE patients may provide valuable insights for early KC diagnosis.^{6,7}

The Corvis ST (Oculus Optikgeräte), which uses ultrahigh-speed Scheimpflug video imaging, captures dynamic corneal response (DCR) parameters during air-puff induced deformation over 30 ms, providing a detailed biomechanical assessment.⁹ However, conventional DCR parameters integrated into the device software primarily

assess anterior surface deformation at specific time points, potentially missing subtle biomechanical abnormalities occurring at other phases or in different corneal regions.¹⁰ Furthermore, these parameters may not fully capture the spatial heterogeneity of corneal biomechanical responses, limiting their sensitivity in detecting early or localized ectatic changes.^{10–15}

Despite encouraging initial reports, the diagnostic performance of the Corvis ST in KC detection remains in the absence of morphological metrics, and even integrated tomographic-biomechanical approaches struggle to identify subclinical disease.^{15–18} Conventional analyses typically assess displacement at a few anterior corneal locations and selected time points. In contrast, pixel-level displacement tracking across the entire corneal area and throughout all video frames has been scarcely reported.^{10,19} Likewise, pixel-level analysis of stromal gray scale patterns and transient corneal thickness variation, that may potentially reveal microstructural and viscoelastic changes, remains largely unexplored.^{11,20} Leveraging the full spatiotemporal profiles from the Corvis ST instrument, may enable a more sensitive characterization of subtle biomechanical changes associated with early KC.

Artificial intelligence (AI), particularly computer vision and machine learning (ML), offers a powerful framework for enhancing KC detection.^{19,21–25} Pixel-level analysis of DCR videos leverages its high spatial (within frames) and temporal (between frames) resolution to identify subtle biomechanical changes that may be undetectable using conventional methods.²⁶ Various supervised ML models, including linear discriminant analysis, decision trees, support vector machines, random forest (RF), and K-nearest neighbors (KNNs), have been investigated for KC detection.²⁷ Unsupervised models, such as principal component analysis (PCA) and K-means clustering, have also been explored in data exploration and feature selection previously.²¹ Notably, ensemble approaches, such as voting, combining multiple base models, have been reported to further enhance diagnostic accuracy.¹⁵

Although large and vision language models have been emerged as state-of-the-art in various vision and ophthalmology applications,^{28–31} they have not yet been applied to KC detection from corneal videos. However, several recent studies, including ours, have explored deep learning approaches, particularly convolutional neural networks, for KC detection using corneal videos. Nevertheless, these efforts have primarily focused on detecting established KC and have shown limited performance in identifying subclinical or early-stage disease.

This study presents a novel computer vision and AI-derived index to enhance the early detection of corneal E by fully leveraging spatiotemporal features of Corvis ST video data. Building on a previously established computer vision-enabled video preprocessing pipeline, we generate 2-dimensional (2D) representations that capture dynamic corneal deformation, stromal gray scale fluctuations, and transient changes in stromal thickness, associated with biomechanical signatures that remain largely unexplored in existing literature. From these comprehensive visual maps, we extract novel, high-dimensional features that are

subsequently integrated into a voting classifier ensemble. This approach enables robust and accurate identification of subclinical and early KCN, outperforming Pentacam and Corvis ST built-in tomographic and biomechanical indices (TBIs) and offering a powerful tool for KC management.

Methods

Subjects and Data

This retrospective multicenter study was conducted at 2 private ophthalmological centers in Iran and Germany. Institutional approval for the retrospective analysis of anonymized patient data was obtained from both centers. The study adhered to the Declaration of Helsinki and its subsequent amendments. Written informed consent was obtained from participants before data use, and ethical approval was obtained from the institutional review board offices of both centers.

Data set 1 comprised clinical data from 314 participants, aged 18 to 45 years, who were examined at the Salouti Eye Center, Iran, between April 2021 and December 2024. Data set 2 included data from 137 participants, aged 18 to 45 years, who presented at the Center for Refractive Surgery Muenster in Germany between March 2021 and September 2024.

All patient data were fully deidentified before analysis in both countries to ensure confidentiality and compliance with data protection regulations.

Definitions and Clinical Diagnosis of Subjects

Normal: participants presenting for LVC with: (1) stable corrected distance visual acuity (CDVA) of 1.0 or better (2 years pre-LVC, 1 year post-LVC), (2) Normal slit lamp biomicroscopy and retinoscopy, (3) topographical KC classification (TKC) grading KC0 and Belin/Ambrósio total deviation value (Belin/Ambrósio Enhanced Ectasia Display [BAD-D]) <1.6 ,^{32,33} and (4) no family history of KC. One eye was selected randomly to minimize bias from bilateral eye correlation.

Very asymmetric ectasia with NT (VAE-NT): participants with contralateral eye diagnosed with frank KC (as defined below) and: (1) CDVA ≥ 1.0 , (2) no KC signs on slit lamp examination, (3) NT (KC percentage index <60 , paracentral inferior-superior asymmetry <1.45 diopters [D], maximum keratometry [Kmax] <47.40 D, thinnest pachymetry ≥ 480 μm , BAD-D <2.6),^{32,34} and (4) with or without posterior corneal elevation at thinnest point of >11 μm .³⁵ Only the eye with NT was analyzed, regardless of tomography findings.

Mild KC (MKC): participants with contralateral eye diagnosed with frank KC (as defined below), and: (1) CDVA 0.8 or better, (2) <2 slit lamp findings and no central scars, (3) TKC grading 1 (Kmax <48.5 D, thinnest pachymetry ≥ 470 μm , inferior-superior asymmetry >1.45 D, BAD-D <4.0 standard deviation [SD]), posterior corneal elevation at thinnest point >11 μm).^{32,34,35} From these participants with clinically evident KC in one eye and MKC in the fellow eye, only the eye with MKC was analyzed.

Frank KC: frank KC (used for group definition) was diagnosed based on the following criteria: (1) CDVA worse than 0.8, (2) one or more slit lamp findings (Fleischer ring, Vogt striae, or apical scarring),³⁶ and (3) severity ≥ 2 in the well-established TKC index from topometric display and tomography imaging in the Pentacam HR system, which is based upon the Amsler–Krumeich classification of KC on Scheimpflug topography,³³ and BAD-D of >2.6 .³²

All cases were blindly reviewed by an independent anterior segment expert (H.A.) to confirm inclusion criteria. Clinical and

tomographic data were also masked and re-evaluated by an expert from another center for further validation. To avoid bias, the N group was matched to the VAE-NT and MKC groups based on age and gender. Participants were instructed to discontinue rigid and soft contact lens wear for at least 4 and 2 weeks, respectively, before examination. Patients with previous ocular surgery (including corneal cross-linking) or disease, systemic disease with ocular implications, and glaucoma or hypotonic therapies were excluded. All participants underwent a comprehensive ophthalmic examination, including Pentacam HR and Corvis ST (Oculus Optikgeräte GmbH) assessments. Corvis ST analysis techniques, DCR parameter definitions, and summary indices were as previously described.¹⁹ Demographic and ophthalmic data, encompassing Pentacam HR corneal morphological parameters and Corvis ST DCR parameters, were exported to a custom spreadsheet and linked to corresponding Corvis ST videos. Each video was assigned a serial number and labeled as N, VAE, or MKC.

Computer Vision Pipeline to Extract Most Relevant Biomechanical Features

A total of 451 high-quality slow-motion Corvis ST videos (13 frames/second) were analyzed. Each video, with a 576×214 pixel resolution, captured the 10-second DCR across an 8.8 mm horizontal corneal segment. The video sequence of corneal deformation from initial convexity to maximum concavity and recovery was captured by a high-speed camera using a 470 nm blue light emitting diode.⁹ The computer vision pipeline included automated video frame processing using the OpenCV library (version 4.5.4, <http://opencv.org>). Building upon our prior computer vision pipeline for DCR analysis,¹⁹ this study expands the methodology to comprehensively extract features from the (1) corneal stroma, focusing on gray scale intensity changes as indicatives for light scattering variations and (2) dynamic corneal thickness fluctuations indicative of underlying viscoelastic properties. The pipeline enabled the simultaneous extraction of 3 key biomechanical parameters including corneal deformation, stromal light scattering characteristics, and dynamic thickness changes from each video, providing a comprehensive corneal biomechanical assessment.

Similar to our previous work,¹⁹ corneal deformation analysis included identifying corneal regions in video frames, binarizing the image, aligning (registering) frames to compensate eye movement, generating a corneal skeleton, and calculating pixel vertical displacement relative to the original corneal shape. This produced a 2D array representing dynamic corneal shape changes throughout the video (Fig 1; panels A and B). The following features were then extracted and quantified for the downstream ML analysis: frame serial number at key corneal positions (first and second applanation and maximum concavity), pixel distance from the unaltered position to each key position per frame, area under the curve (AUC) per frame and key position, and AUC differences and sums between key positions.

Corneal stromal gray scale and optical flow analysis³⁷ included flattening corneal epithelium in each frame, followed by underlying corneal segment realignment. The mean pixel intensity was calculated for each column, generating a 1-dimensional profile representing average stromal intensity across the corneal width per frame. These 1-dimensional profiles were sequentially aligned into a 2D array, where rows represent frame-wise average stromal intensity and columns represent intensity changes at specific corneal meridians (Fig 1; panels C and D). The following features were then extracted and quantified: overall peak intensity, time of intensity rise, peak, and fall, metrics of the

transient, oblique, high-intensity gray scale feature (onset frame, propagation speed, acceleration, deceleration) based on the optical flow analysis.

Corneal thickness analysis was performed using the previously flattened corneal segment, which included the epithelium. A spline curve was applied to smooth the endothelial boundary and reduce noise, while removing outlier epithelial pixels (potentially representing tear film debris). Corneal thickness was then calculated column-wise, producing a 1-dimensional thickness profile for each video frame. These sequential profiles were aligned to form a 2D array, capturing spatiotemporal thickness variations across the cornea (Fig 1, panels E and F). Row and column positions of minimum mean thickness, their AUC, and excess kurtosis were extracted and quantified for the downstream analysis.

To further investigate regional variations, interregional and intraregional correlations were examined by comparing the 2 halves of the corneal profile. Lagged cross-correlations between the 2 halves of each array were computed per row, and correlation patterns were summarized using symmetric autocorrelation (reversed array), intersegmental cross-correlation at lag 0 (reversed half), and the average Pearson correlation coefficient (excluding near-zero variance signals). Summary statistics (mean, SD, median, interquartile range) were calculated for each correlation measure across all time points/samples. Additionally, the mean of cross-correlation values exceeding a defined threshold was computed for each array (dynamics, intensities, and thickness) to quantify deviations from symmetrical patterns (mean above-threshold correlation). Normalized lagged cross-correlation heatmaps, illustrating half similarity, were generated. This analysis quantified biomechanical corneal response similarity, dissimilarity, and variability.

Data Set Splitting

Data set 1 (Iran, training and testing) was further randomly divided into training (67%) and testing (33%) subsets. Data set 2 (Germany) was used for external validation only. Recognizing that VAE-NT and MKC represent early E stages,^{3,38} we combined them to facilitate feature space visualization³⁹ as well as enabling binary classification of N versus E.

Feature Extraction and Selection

To extract meaningful features from the corneal deformation, stromal gray scale intensity, and corneal thickness variation 2D arrays, descriptive statistics and regional correlations were quantified. This process generated an initial set of 94 features. Feature cleaning and preprocessing were then performed to remove attributes with low relevance based on: >30% identical values, >30% missing values, or variance <0.07. Multicollinearity was also assessed using variance inflation factor analysis, and features with a variance inflation factor >5 were removed. The remaining 66 features were then subjected to recursive feature elimination with a RF estimator on the training subset to identify the most informative feature subset for model development. This automated process identified the most informative features for model development.

Developing the AI-Derived Biomechanical Index

To develop the biomechanical index, a soft voting ensemble classifier was trained using the selected features, combining probability predictions from 3 RF models (optimized via grid search) and a support vector machine classifier. Soft voting enhances both confidence/robustness and accuracy.¹⁵ The optimal

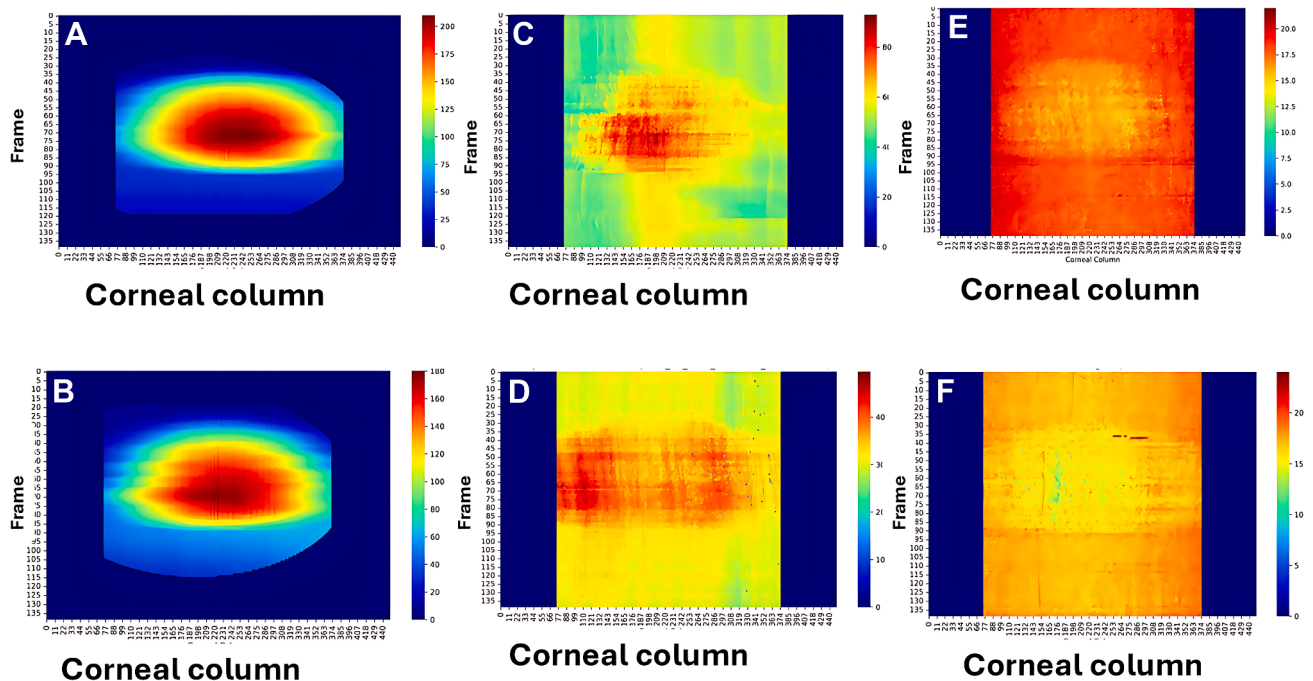


Figure 1. Example of 2D feature arrays: (A, B) corneal movement (dynamics), (C, D) gray scale intensity fluctuations, and (E, F) corneal thickness variations. x-axis: corneal pixel columns; y-axis: video frames. Color scale represents: (A, B) pixel position change, (C, D) average column pixel intensity, and (E, F) thickness change (pixels). 2D = 2-dimensional.

cutoff, separating N from E, was determined using the Youden index.⁴⁰

Evaluation Methods

The identified biomechanical index was validated based on several approaches. First, we evaluated the diagnostic performance of the AI-derived biomechanical index based on quantitative measures including AUC, sensitivity, specificity, and F1 score. Second, we compared the performance of the AI-derived biomedical index with the existing Pentacam and Corvis ST indices. Third, we conducted several robustness and ablation evaluation studies. Specifically, we assessed the importance of selected features based on linear discriminant analysis and RF using the training/testing data set (ie, wrapping classifiers). Additionally, we conducted redundancy assessment by investigating the relationship among the AI-derived index, selected features, and existing Corvis and Pentacam indices based on Spearman's rank correlation. Fourth, we investigated the explainability by plotting KNN decision boundaries for 3-class classification (N, VAE-NT, and MKC) using the training/testing data set and visualized the distribution and descriptive statistics for each feature. Fifth, we validated features based on unsupervised learning approaches. More specifically, principal component regression analysis was used to identify major principal coefficients. The transformed features were then scaled and normalized to lie between 0 and 1, which served as an unsupervised assessment of initial features. A cutoff point on this scale was determined using the Youden index, calculated from the performance of the linear regression model in discriminating eyes between N and E groups based on the training subset. The distribution of the features based on the testing subset was then visualized using these principal components to ensure robustness. K-means clustering further examined data and visualized grouping patterns using multiple principal components.

Additionally, a supervised learning technique was used to plot decision boundaries for the N, VAE-NT, and MKC, demonstrating the feature set's ability to delineate distinct classes by providing a visualization of decision boundaries between groups in the reduced feature space. These analyses collectively provided insights into the separability of the classes and the overall data structure based on the identified features. Finally, we evaluated the importance of the selected features and validated findings based on cross-center (external) validation.

Statistical Analysis

Statistical analyses were performed in Python (version 3.9.13).^{41–43} Normally distributed data are presented as mean $\pm$ SD. The Shapiro–Wilk test was used to assess the normality of continuous variables. Independent samples t-tests and Wilcoxon rank-sum tests were used for normally and non-normally distributed data, respectively (significance level: $P < 0.05$).

A soft voting ensemble classifier, combining 3 RF models and a support vector machine, was trained for classification. Unsupervised analysis (PCA and k-means clustering) explored data structure and validated feature selection. A subsidiary scale, derived from PCA and linear regression, was created with a Youden index—determined cutoff. K-nearest neighbors visualized decision boundaries in PCA-reduced space. Area under the receiver operating characteristic curve was calculated for performance assessment. Pairwise AUC comparisons were performed using DeLong's nonparametric approach.⁴⁰ Additionally, separation curves based on normalized cutoffs (Z-transformation) were used to assess cutoff criteria tolerance.⁴⁴ The area under the separation curve (AUSEP), calculated between 50% and 100% accuracy and ± 1 SD, was also used to evaluate discriminatory ability. Higher AUSEP

values indicate greater tolerance and better discrimination. The Youden index determined optimal cutoffs. Bootstrapping (1000 resamples) assessed the statistical significance of AUSEP differences. Spearman rank correlation analyzed variable correlations. All computations were performed on a single graphics processing unit.

A power analysis ($\alpha = 0.05$, power = 0.8) determined a required sample size of 117 participants per group (1-tailed t-test, effect size = 0.33). Accounting for 5% data loss, 118 participants per group (236 total) were targeted. The achieved power for the final sample sizes (E: 142, N: 161) ranged from 0.80 to 0.88 (effect size 0.30–0.33).

Results

Demographics

The first data set, collected from Iran, included 314 eyes: 167 from N, 83 with VAE-NT, and 64 with MKC. There were no significant differences in age ($P = 0.198$) or gender distribution ($P = 0.887$) across the groups. However, significant differences were observed in Kmax, BAD-D, Corvis Biomechanical Index (CBI), and TBI, with all P values < 0.001 . The second data set, from Germany, comprised 137 eyes: 68 from N, 33 with VAE-NT, and 36 with MKC. Age ($P = 0.441$) and gender ($P = 0.940$) were comparable among groups, whereas Kmax, BAD-D, CBI, and TBI again differed significantly ($P < 0.001$) (Tables 1 and 2).

Video Preprocessing, Feature Extraction, and Selection

A total of 15 videos were excluded because of processing failures, 11 from data set 1 (6 N, 2 VAE-NT, 3 MKC) and 4 from data set 2 (2 N, 2 VAE-NT) leading to 436 videos (303 from data set 1, 133 from data set 2) for the downstream analysis. Data set 1 was further split into training (107 N, 95 E) and testing (54 N, 47 E) subsets.

Feature selection began with 94 features. After preprocessing to remove low-variance, redundant, and incomplete variables, 66 features remained. The remaining 66 features were then subjected to recursive feature elimination (RFE) with an RF estimator on the training subset to identify the most informative feature subset for model development. Finally, RFE with a RF estimator identified an optimal 10-feature subset. The definitions of these features are detailed in Table S1 (available at www.ophtalmologyscience.org).

Individual video preprocessing (noise reduction, artifact removal, 2D array development, feature extraction) took 47.5 s (central processing unit) and 47.6 s (wall).

AI-Derived Biomechanical Index

The AI-derived index, based on the voting classifier model, achieved an accuracy of 96%, an AUC of 0.995, and a maximum Youden J index of 0.951 at a probability threshold of 0.478. At this cutoff, sensitivity, specificity, precision, and F1 score for the N group were 0.97, 0.97, 0.96, and 0.96, and for the E group were 0.96, 0.96, 0.97,

and 0.96, respectively (Fig 2). Model inference wall time was approximately 50 ms.

Performance Based on the Cross-Center Validation

The AI-derived biomechanical index demonstrated a perfect discrimination with an AUC of 1.0, followed by TBI and BAD-D with AUCs of 0.946 and 0.940, respectively (Table 3). The best-performing single biomechanical parameters were delta arc length of the corneal surface in first applanation (mm) and first applanation time with AUCs of 0.76 and 0.592, respectively. A comprehensive overview of the performance of other evaluated parameters is provided in Table S2 (available at www.ophtalmologyscience.org).

Although the AI-derived biomechanical index had a higher AUC compared to TBI, BAD-D, CBI, or maximum pachymetry progression index, the difference was not statistically significant ($P > 0.059$), as detailed in Table S3 (available at www.ophtalmologyscience.org). However, in terms of AUSEP, the AI-derived biomechanical score outperformed all individual and summary parameters with a P value of < 0.001 , highlighting its superior ability to separate the classes robustly (Table S4, available at www.ophtalmologyscience.org).

Figures 3 and 4 show the receiver operating characteristic curves, AUSEP curves, and confusion matrices for the novel index and existing instrument indices.

Unsupervised Principal Component Regression and K-Means Clustering

To further validate the high performance of the voting classifier and independently confirm the discriminative power and underlying data structure revealed by the supervised model, we employed unsupervised learning techniques (PCA and k-means clustering).

Principal component analysis was used to visualize the data distribution in a reduced-dimensional space. Figure 5 illustrates 3-dimensional PCA plots of the training and testing data sets, demonstrating clear separation based on selected principal components and a PC regression normalized scale. A cutoff value of 0.442 (based on Youden J index) from linear regression effectively separated N from E cases. This separation was further validated in the external validation data set, as shown in Figure S1 (available at www.ophtalmologyscience.org), in which new, unseen cases of VAE-NT were distinctly separated from N corneas. This demonstrates the robustness of the features and the inherent separability of the data.

K-means clustering further examined data grouping patterns in this reduced space. Figures S2 to S4 (available at www.ophtalmologyscience.org) illustrate the distinct separation of the N and E groups using the training and testing data sets, whereas Figure S5 (available at www.ophtalmologyscience.org) shows this same separation for the independent external validation data set. This unsupervised analysis confirms that the data naturally form clusters corresponding to the defined classes.

Table 1. Demographic and Ocular Characteristics of the Subjects in Data Set 1 (Training/Testing)

Parameter	Normal	VAE-NT	MKC	P
N	167	83	64	NA
Eye (right/left)	90/77	44/39	31/33	0.812
Gender (male/female)	86/81	39/44	34/30	0.887
Age (years)				0.198
Median	25	24	25	
IQR	5.0	4.0	4.0	
Range	18–48	19–45	18–47	
IOP (mmHg)				0.011
Median	16	15	16	
IQR	2.0	3.0	3.0	
Range	10–22	11–19	10–21	
Kmax (D)				0.041
Median	43.7	44.0	45.60	
IQR	1.4	1.3	2.5	
Range	39.8–47.2	38.7–47.3	42.9–48.4	
CCT (μm)				<0.001
Mean $\pm$ SD	521 $\pm$ 30.22	519 $\pm$ 20.13	498 $\pm$ 16.13	
Range	477–675	483–638	470–543	
C. Vol D 3 mm				0.911
Mean $\pm$ SD	3.88 $\pm$ 21.12	3.74 $\pm$ 16.33	3.53 $\pm$ 18.87	
Range	3.50–4.16	3.52–4.12	3.47–3.98	
BAD-D				<0.001
Median	0.61	1.39	2.68	
IQR	0.42	0.82	0.52	
Range	−0.04–1.58	−0.18–2.48	1.68–3.88	
A1 velocity [m/s]				0.044
Median	0.15	0.16	0.17	
IQR	0.04	0.03	0.09	
Range	0.13–0.18	0.12 to −0.19	0.12–0.22	
A1 time [ms]				0.008
Median	7.33	7.28	7.19	
IQR	0.09	0.13	0.11	
Range	7.09–7.56	6.94 to −7.41	6.90–7.40	
A2 velocity [m/s]				0.048
Median	−0.28	−0.28	−0.29	
IQR	0.2	0.2	0.3	
Range	−0.31 to −0.23	−0.33 to −0.24	−0.35 to −0.23	
A2 Defl. Amp [mm]				0.038
Median	0.09	0.11	0.11	
IQR	0.01	0.01	0.01	
Range	0.08–0.10	0.09–0.13	0.09–0.13	
A2 Defl. length [mm]				0.066
Median	2.88	2.89	2.83	
IQR	0.14	0.19	0.21	
Range	2.54–3.31	2.56–3.40	2.44–3.24	
DA ratio max (2 mm)				0.059
Median	4.50	4.49	4.58	
IQR	0.32	0.32	0.22	
Range	3.13–4.79	3.23–4.82	3.88–5.23	
PachySlope (μm)				0.008
Median	42.28	44.61	49.81	
IQR	10.22	12.88	15.88	
Range	20.50–60.25	30.77–86.78	33.05–90.08	
ARTh				0.021
Median	461.10	418.32	384.22	
IQR	64.22	50.87	45.43	
Range	425.22–554.45	320.27–483.25	302.27–460.16	
biOP (mmHg)				0.005
Median	16.8	14.8	15.5	
IQR	2.9	3.4	3.5	
Range	10.3–22.0	11.2–21.2	11.5–21.3	

Table 1. (Continued.)

Parameter	Normal	VAE-NT	MKC	P
SP A1				0.008
Median	106.04	99.19	84.19	
IQR	10.02	9.67	10.67	
Range	91.30–129.68	82.35–111.36	69.35–100.36	
CBI				<0.001
Median	0.03	0.22	0.68	
IQR	0.02	0.47	0.26	
Range	0.00–0.81	0.01–1.00	0.00–1.00	
TBI				<0.001
Median	0.03	0.31	0.94	
IQR	0.08	0.52	0.02	
Range	0.00–0.74	0.00–1.00	0.88–1.00	

A1 = first applanation; A1 time = time from the measurement beginning to the A1; A1 velocity = velocity of the corneal apex during A1; A2 = second applanation; A2 Defl. Amp. = corneal deflection amplitude during A2; A2 Defl. Length = Horizontal length of the flattened cornea at the A2; A2 velocity = velocity of the corneal apex during A2; ARTh = Ambrósio relational thickness (horizontal); BAD-D = Belin/Ambrósio enhanced ectasia total deviation index; bIOP = biomechanical intraocular pressure; C. Vol D 3 mm = corneal volumes at 3 mm; CBI = Corvis biomechanical index; CCT = central corneal thickness; DA = deflection amplitude; DA ratio max (2 mm) = Apex MaxDA/MaxDA at 2 mm from the apex; IOP = intraocular pressure; IQR = interquartile range (outlier-excluded); Kmax = maximal apical keratometry; MKC = mild KC; N = number of subjects; NA = not applicable; SP A1 = stiffness parameter at first applanation; SD = standard deviation; TBI = tomographic and biomechanical index; VAE-NT = VAE with NT.

Bold type signifies $P \leq 0.05$.

Additionally, a supervised learning technique was used to plot decision boundaries. Figure S6 (available at www.ophtalmologyscience.org) illustrates KNN decision boundaries in a PCA-reduced space, visually confirming the clear separation between N eyes and the E groups, as well as the significant overlap between VAE-NT and MKC.

Collectively, these unsupervised analyses confirm that the feature set's ability to delineate distinct classes is based on an inherent, fundamental difference in the data structure, thereby independently validating the high performance of our supervised model and mitigating concerns about overfitting.

Feature Ranking

Random forest and linear discriminant analysis showed “mean above-threshold correlation of dynamics” as the most important feature, followed by “mean intersegmental lagged cross-correlation for dynamics,” albeit with a performance gap. The feature importance scores from both models, presented in Figure S7 (available at www.ophtalmologyscience.org), confirm the relative contribution of each of the 10 features.

The superior performance of “mean above-threshold correlation of dynamics” was further supported by the receiver operating characteristic curve analysis, which yielded an AUC of 0.87 when classifying VAE-NT versus N cases from the external validation data set (Fig S8, available at www.ophtalmologyscience.org). Furthermore, the combined discriminative power of the top 2 features, “mean above-threshold correlation of dynamics” and “mean intersegmental lagged cross-correlation for dynamics,” was visually confirmed through a KNN decision boundary plot (Fig S9, available at www.ophtalmologyscience.org). This plot demonstrates a clear and wide separation between the VAE-NT versus N

cases, while also showing the significant overlap between VAE-NT and MKC cases. Finally, Figure S10 (available at www.ophtalmologyscience.org) provides kernel density plots of selected features, which visually demonstrate the separation between the N and VAE-NT groups, thereby supporting the findings of the receiver operating characteristic curve analysis.

Correlation Analysis

Spearman rank correlation analysis investigated the relationship between the novel voting score, our selected features, and established tomographic and biomechanical parameters. The results of this extensive analysis are presented in a series of figures for clarity. We first examined the relationship between our AI-derived index and existing parameters. Figure S11 (available at www.ophtalmologyscience.org) displays the correlations with tomographic indices, whereas Figure S12 (available at www.ophtalmologyscience.org) shows correlations with biomechanical indices. These heatmaps present correlation coefficients and visually indicate statistical significance. Figure S13 (available at www.ophtalmologyscience.org) provides a more detailed view, illustrating the correlations between our individual novel features, the voting score, and established parameters.

To explore correlations within the most challenging subset of cases, we focused on samples with a low BAD-D value (BAD-D < 1.6 SDs). Figures S14 and S15 (available at www.ophtalmologyscience.org) present the correlations of the novel score with TBIs, respectively, within this specific subgroup.

Notably, “mean above-threshold correlation of dynamics,” despite its high classification importance for VAE detection, showed no significant correlation with any single parameter or multivariable summary index. This suggests that this feature captures unique, previously unmeasured

Table 2. Demographic and Ocular Characteristics of the Subjects in Data Set 2 (External Validation)

Parameter	Normal	VAE-NT	MKC	P
N	68	33	36	NA
Eye (right/left)	32/36	15/18	19/17	0.908
Gender (male/female)	35/33	16/17	19/17	0.940
Age (yrs)				0.441
Median	25	25	24	
IQR	6.0	6.0	5.0	
Range	18–48	18–46	20–47	
IOP (mmHg)				0.053
Mean $\pm$ SD	15 $\pm$ 2.0	15 $\pm$ 3.0	16 $\pm$ 2.0	
Range	11–22	10–19	12–21	
Kmax (D)				0.030
Median	43.3	43.0	44.9	
IQR	1.5	1.4	2.7	
Range	39.7–47.2	39.5–47.3	42.1–48.2	
CCT (μ m)				<0.001
Mean $\pm$ SD	533 $\pm$ 22.22	529 $\pm$ 20.13	495 $\pm$ 16.12	
Range	485–652	491–641	473–523	
C. Vol D 3 mm				0.989
Mean $\pm$ SD	3.75 $\pm$ 21.12	3.76 $\pm$ 16.30	3.68 $\pm$ 18.67	
Range	3.54–4.04	3.59–4.11	3.40–3.89	
BAD-D				<0.001
Median	0.50	1.25	2.68	
IQR	0.38	0.77	0.44	
Range	−0.06 to 1.55	−0.08 to 2.40	1.70–3.61	
A1 velocity [m/s]				0.038
Median	0.15	0.16	0.17	
IQR	0.05	0.07	0.06	
Range	0.11 to −0.19	0.11 to −0.18	0.13–0.25	
A1 time [ms]				0.002
Median	7.38	7.32	7.19	
IQR	0.05	0.13	0.09	
Range	7.19–7.66	7.05 to −7.37	7.00–7.35	
A2 velocity [m/s]				0.058
Median	−0.28	−0.28	−0.29	
IQR	0.2	0.3	0.2	
Range	−0.32 to −0.25	−0.32 to −0.25	−0.33 to −0.24	
A2 Defl. Amp [mm]				0.050
Median	0.09	0.11	0.11	
IQR	0.01	0.01	0.02	
Range	0.08–0.11	0.08–0.12	0.09–0.14	
A2 Defl. Length [mm]				0.071
Median	2.89	2.88	2.84	
IQR	0.12	0.16	0.20	
Range	2.61–3.28	2.48–3.33	2.50–3.20	
DA ratio max (2 mm)				0.061
Median	4.54	4.49	4.56	
IQR	0.19	0.23	0.22	
Range	3.10–4.75	3.48–4.89	3.90–5.12	
PachySlope (μ m)				0.006
Median	42.30	44.67	48.00	
IQR	10.22	10.88	8.88	
Range	28.44,66.44	32.77,76.44	38.05,83.12	
ARTh				<0.001
Median	450.22	402.32	371.83	
IQR	34.22	40.87	55.04	
Range	405.22–534.45	333.27–490.43	289.27–460.43	
biOP (mmHg)				0.147
Mean $\pm$ SD	15 $\pm$ 3.2	15 $\pm$ 3.0	16 $\pm$ 2.8	
Range	10–22	10–22	12–21	
SP A1				0.002
Median	103.04	94.19	81.19	
IQR	8.44	11.54	13.30	
Range	88.46–118.32	71.35–110.42	67.35–98.36	

Table 2. (Continued.)

Parameter	Normal	VAE-NT	MKC	P
CBI				<0.001
Median	0.06	0.32	0.60	
IQR	0.04	0.37	0.29	
Range	0.00–0.72	0.01–1.00	0.00–1.00	
TBI				<0.001
Median	0.02	0.28	0.98	
IQR	0.02	0.60	0.01	
Range	0.00–0.61	0.05–1.00	0.80–1.00	

A1 = first appplanation; A1 time = time from the measurement beginning to the A1; A1 velocity = velocity of the corneal apex during A1; A2 = second appplanation; A2 Defl. Amp. = corneal deflection amplitude during A2; A2 Defl. Length = horizontal length of the flattened cornea at the A2; A2 velocity = velocity of the corneal apex during A2; ARTh = Ambrósio relational thickness (horizontal); BAD-D = Belin/Ambrósio enhanced ectasia total deviation index; bIOP = biomechanical intraocular pressure; C. Vol D 3 mm = corneal volumes at 3 mm; CBI = Corvis biomechanical index; CCT = central corneal thickness; DA = deflection amplitude; DA ratio max (2 mm) = Apex MaxDA/MaxDA at 2 mm from the apex; IOP = intraocular pressure; IQR = interquartile range (outlier-excluded); Kmax = maximal apical keratometry; MKC = mild KC; N = number of subjects; NA = not applicable; SP A1 = stiffness parameter at first appplanation; SD = standard deviation; TBI = tomographic and biomechanical index; VAE-NT = VAE with NT.

Bold type signifies $P \leq 0.05$.

aspects of corneal behavior, highlighting its potential to provide complementary diagnostic information.

Furthermore, a significant negative linear relationship was observed between minimum pachymetry and the voting score in N corneas, even with a wide central corneal thickness (CCT) range. Figure S16 (available at www.opthalmologyscience.org) visually demonstrates this correlation, which indicates that the voting score reflects independent structural or biomechanical characteristics beyond CCT. This persistent correlation within the N group suggests that the voting score captures subtle corneal structure variations even in the absence of E.

Discussions

This study presents a novel AI-derived biomechanics index that enhances early detection of corneal E by leveraging spatiotemporal features from Corvis ST videos. Through

detailed pixel-level analysis of corneal deformation, stromal gray scale fluctuations, and dynamic thickness changes, the index captures viscoelastic abnormalities often undetectable by traditional TBIs. The index was derived using a combination of supervised and unsupervised ML models to detect subtle viscoelastic abnormalities even in challenging subclinical cases before morphological changes became apparent. Our findings represent a significant advancement in the paradigm of early E detection, challenging the current reliance on structural parameters alone for early E screening and advocating for the integration of dynamic biomechanical assessment into routine diagnostics.

We extracted a comprehensive set of spatiotemporal features and evaluated them through multiple analytical approaches. Unlike Wang et al,^{14,15} who used Digital Image Correlation to track deformation by analyzing displacement and strain from segmented corneal regions, our method assessed dynamic corneal behavior through pixel-level

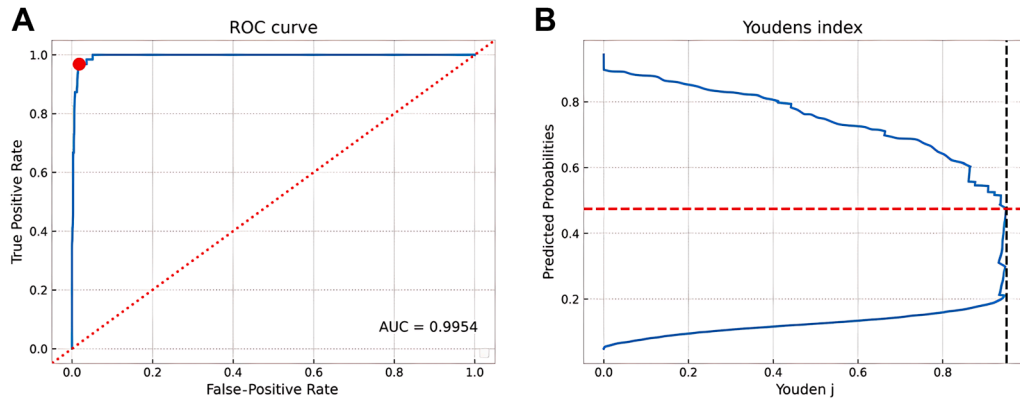


Figure 2. Performance of the AI-derived biomechanical index for binary classification (N vs. E) based on the testing subset. **A**, Receiver operating characteristic curve with AUC of 0.9954. The red dot (upper left corner) indicates the point corresponding to the optimal classification. **B**, Predicted probability scores plotted against Youden J index. The red horizontal dashed line represents the optimal cutoff threshold (0.478), and the vertical blue dashed line indicates the maximum Youden J index (0.942). AI = artificial intelligence; AUC = area under the curve; E = ectasia; N = normal; ROC = receiver operating characteristic.

Table 3. Performance Evaluation of the Top 10 Best-Performing Parameters Using ROC and AUSEP Metrics Based on the External Data Set

Parameter	Mean AUC $\pm$ SE	Sensitivity	Specificity	Accuracy	Youden J	AUSEP
Voting score	1.000 $\pm$ 0.000	1.000	1.000	1.000	1.000	3.976
TBI	0.946 $\pm$ 0.036	0.941	0.913	0.925	0.898	3.682
BAD D:	0.940 $\pm$ 0.038	0.941	0.826	0.875	0.839	3.679
CBI	0.922 $\pm$ 0.042	0.824	0.826	0.825	0.765	3.682
Pachy Prog Index Min.	0.917 $\pm$ 0.044	0.882	0.739	0.800	0.780	3.669
Pachy Prog Index Max.	0.916 $\pm$ 0.044	0.824	0.826	0.825	0.765	3.682
A1 dArc Length [mm]	0.760 $\pm$ 0.068	0.706	0.696	0.7	0.404	3.436
Pachy Min Pos Y:	0.661 $\pm$ 0.075	0.824	0.522	0.65	0.345	2.499
IS-Value:	0.639 $\pm$ 0.076	0.706	0.391	0.525	0.371	3.437
A1 time [ms]	0.592 $\pm$ 0.078	0.647	0.565	0.6	0.269	2.439

A1 = first applanation; A1 dArc Length [mm] = delta arc length of corneal surface in A1; A1 time = time from the measurement beginning to the A1; AUC = area under the receiver operating characteristic curve; AUSEP = area under the separation curve; BAD-D = Belin/Ambrósio Enhanced Ectasia Display; CBI = Corvis biomechanical index; IS-Value = Inferior-superior value; Mean AUC $\pm$ SE = AUC with standard error; MKC = mild KC; Pachy Prog Index Min. = minimum pachymetric progression index; Pachy Prog Index Max. = maximum pachymetric progression index; Score = voting classifier score; TBI = tomographic and biomechanical index.

The table is sorted by mean AUC in descending order.

displacement relative to the original corneal shape, stromal gray scale fluctuations using optical flow, and thickness variations via spline-smoothed corneal segments. Additionally, rather than relying on basic summary statistics like maximum or average values, we performed correlation-based analyses across full 2D arrays and applied RFE for optimal feature selection. Another study¹¹ modeled light intensity distributions using a Gamma function across deformation phases, capturing microstructural changes. However, their histogram-based method excluded epithelial data and did not integrate thickness or displacement metrics. In contrast, our framework combined

biomechanical and structural signals, offering a more holistic view of early ectatic changes. Together, these comparisons highlight the added diagnostic value of our integrated, multimodal analysis.

Both our study and that of Chen et al¹⁰ aimed to advance AI-driven detection of early KC. However, the methodologies differed substantially. Chen et al focused on localized biomechanical features derived from spline-fitted contour parameters, whereas our approach captured broader corneal dynamics through 2D arrays and regional correlation analysis across space and time. Additionally, their ensemble model incorporated 10 classifiers without feature selection,

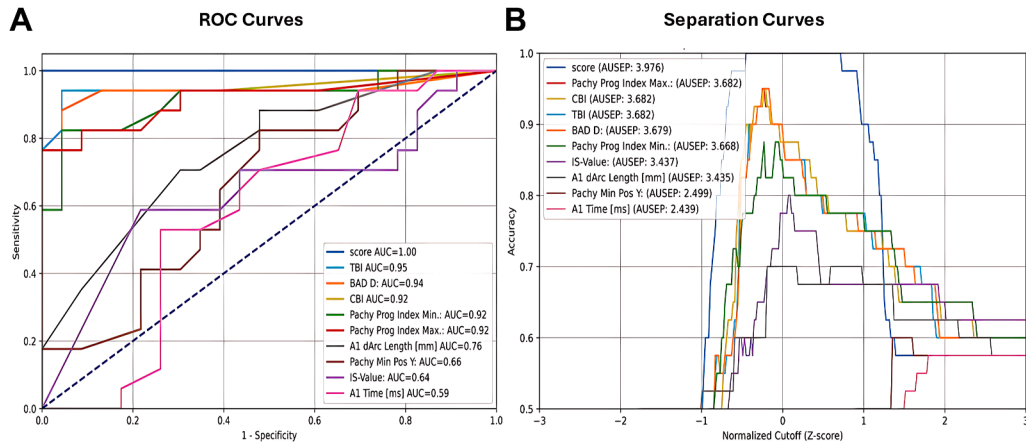


Figure 3. The ROC and SEP curves for binary classification of N eyes vs. eyes with E based on the external validation data set (data set 2). A1 = first applanation; A1 Time [ms] = time from the measurement beginning to A1; A1 dArc Length (mm) = delta arc length of the corneal surface in A1; AUC = area under the curve; BAD-D = Belin/Ambrósio Enhanced Ectasia Display; CBI = Corvis biomechanical index; E = ectasia; IS-Value = Inferior-superior value; N = normal; Pachy Min Pos Y = Y coordinate of minimum corneal thickness; Pachy Prog Index Max. = Maximum pachymetric progression index; Pachy Prog Index Min. = minimum pachymetric progression index; ROC = receiver operating characteristic; score = AI-derived biomechanical score (index); SEP = separation; TBI = tomographic biomechanical index; AUC = area under the curve.

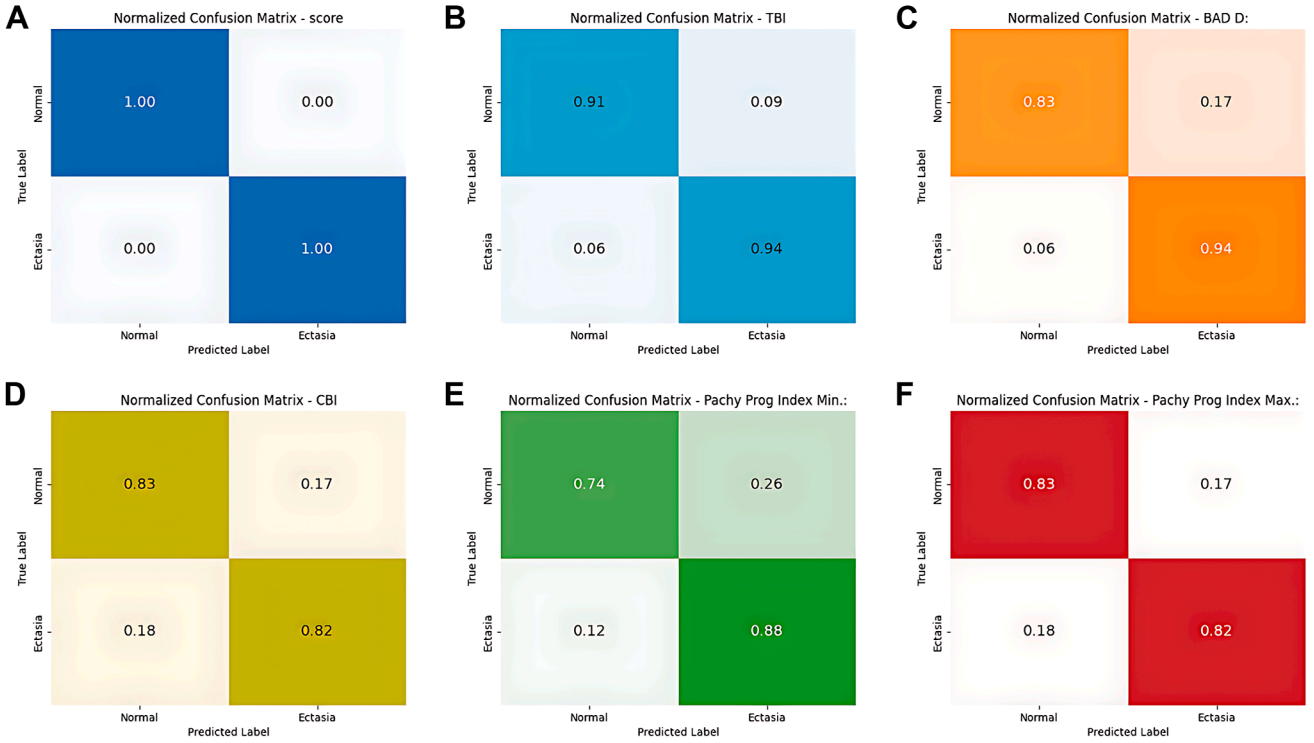


Figure 4. Normalized confusion matrices based on the external validation data set for binary classification of N vs E. Ectasia group is composed of VAE (VAE with normal topography) and MKC. **A**, AI-derived biomechanical score (index), **(B)** TBI, **(C)** BAD-D, **(D)** Corvis biomechanical index, **(E)** Pachy Prog Index Min, and **(F)** Pachy Prog Index Max. AUSEP = area under the separation curve; BAD-D = Belin/Ambrósio Enhanced Ectasia Display; CBI = Corvis biomechanical index; E = ectasia; MKC = mild KC; VAE = very asymmetric ectasia; N = normal; Pachy Prog Index Max. = maximum pachymetric progression index; Pachy Prog Index Min. = minimum pachymetric progression index; TBI = tomographic biomechanical index.

whereas our optimized ensemble was built upon rigorously selected spatiotemporal features, reflecting a more comprehensive biomechanical assessment.

Okon et al²⁰ examined changes in CCT during air-puff deformation and found increased CCT at highest concavity, suggesting corneal incompressibility. Although informative, our approach offers a more advanced and comprehensive assessment of biomechanical behavior. Understanding how CCT relates to novel Corvis ST-derived features is critical, given its influence on corneal mechanics. To ensure the validity of our findings, we actively controlled for confounding factors known to affect biomechanical strength. Patients with ocular or systemic diseases, or a history of ocular surgery, including corneal cross-linking, were excluded. Participants also discontinued contact lens wear before examination to minimize the influence of corneal edema and hydration on the results.

Notably, we found a significant negative correlation between our index and minimum pachymetry ($r = -0.51$, $P < 0.001$) in N corneas, indicating that thinner corneas exhibit distinct biomechanical responses even in the absence of E. This underscores the importance of accounting for CCT in interpreting model outputs and improving diagnostic precision.

Several studies have demonstrated the value of combining tomographic and biomechanical parameters to improve E detection. Song et al¹⁸ showed that a

classification and regression tree model integrating both data types outperformed individual parameters in identifying subclinical KC. Similarly, Ambrósio Jr et al,⁷ through a large multicenter study, introduced the TBIv2 model, which applies advanced AI techniques to enhance diagnostic performance, particularly in detecting VAE-NT cases. Although powerful, TBIv2 and similar indices remain heavily reliant on morphology-driven features, potentially limiting sensitivity to the earliest biomechanical changes. Other studies have emphasized that biomechanical abnormalities often precede tomographic alterations. Liu et al¹⁶ highlighted the superior sensitivity of CBI over BAD-D in detecting subclinical KC, whereas Wu et al⁴⁵ and Lopes et al⁴⁶ acknowledged the limitations of structural features alone, even in integrated indices like TBI. They also noted that models trained predominantly on advanced KC cases may miss earlier phenotypes. In contrast, our index utilized a comprehensive characteristic of spatiotemporal features and was trained on eyes with subclinical VAE and MKC (stage 1 TKC) to enhance sensitivity to early changes. Our findings suggest that integrating detailed biomechanical features with early-stage structural clues provides a more sensitive and clinically relevant framework for E detection.

Research indicates that subtle morphological changes, particularly in central corneal pachymetry distribution and posterior corneal surface alterations, can be detected in

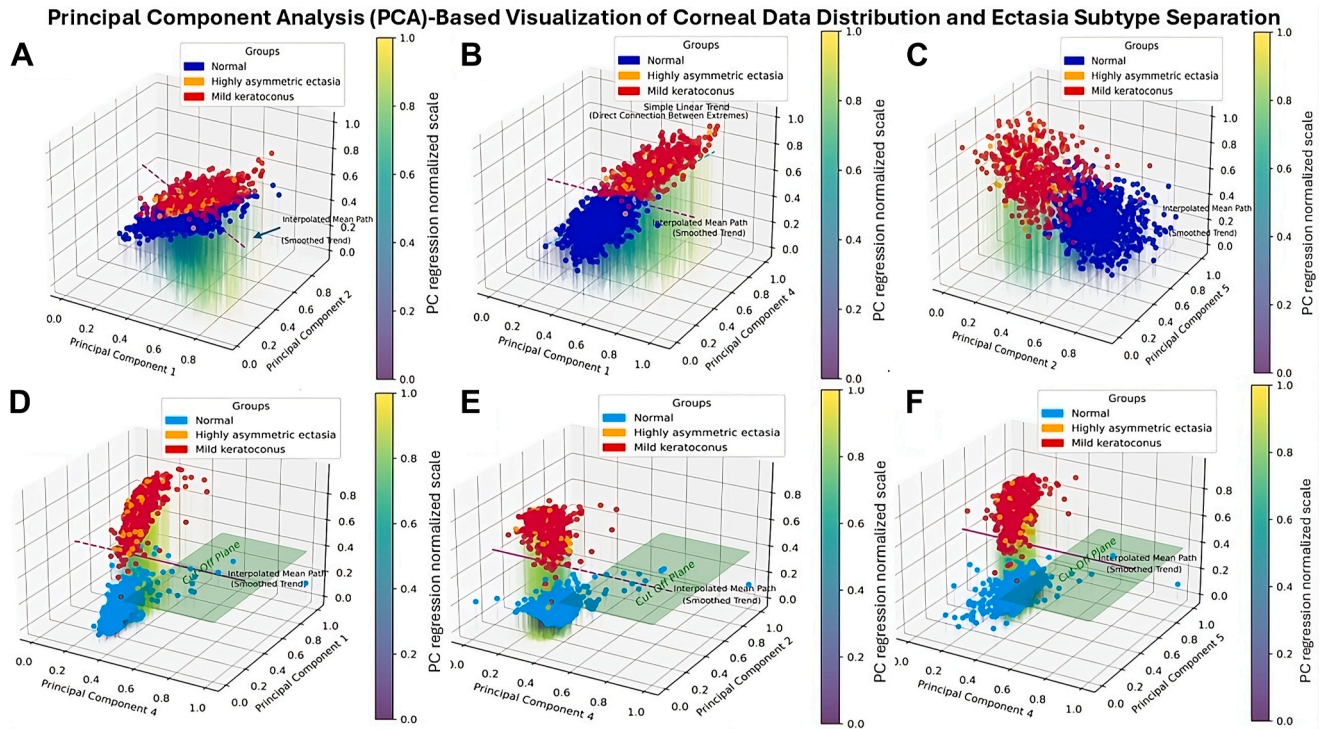


Figure 5. Visualization of features based on the PCA for N, VAE-NT, and MKC groups of the combined training and testing data sets. Panels A–C: 3D PCA plots based on the combined training and testing subsets, (A) PC1 vs. PC2, (B) PC1 vs. PC4, and (C) PC2 vs. PC5. Panels D–F: 3D PCA plots based on the combined testing and external validation subsets. Axes represent normalized eigenvalues of: (D) PC4 vs. PC1, (E) PC4 vs. PC2, and (F) PC4 vs. PC5. The z-axis represents the PC regression normalized scale, with colored lines connecting data points to their respective scale value. The interpolated line indicates the trend of the data distribution. MKC = mild KC; PC = principal component; PCA = principal component analysis; VAE-NT = VAE with NT.

early KC, even with N biomechanics or before anterior surface changes.^{18,47} These findings underscore the importance of incorporating both structural and biomechanical information for early detection, acknowledging that biomechanical abnormalities often precede morphological manifestations. Therefore, these studies collectively underscore the potential of a combined structural and biomechanical approach for early KC detection, while also emphasizing the need for further research to refine the integration and interpretation of these parameters.

Like Herber et al,⁸ our study emphasizes the role of biomechanical profiles in early E detection; however, we focused on localized, pixel-level features rather than global indices such as CBI. Peyman et al⁴⁸ demonstrated that pure biomechanical parameters like CBI can reveal ectatic susceptibility in morphologically N corneas, highlighting their potential in early diagnosis. In contrast, Huo et al⁴⁹ reported that summary indices like TBI and CBI were less effective in cases with subclinical KC, suggesting limited sensitivity to subtle biomechanical abnormalities. Their findings supported the use of specific parameters such as “delta arc length of the corneal surface in A1” for improved early detection. These observations highlight the importance of selecting

appropriate clinical populations and developing relevant indices for early-stage E.

Zhang et al⁵⁰ highlighted that N tomographic findings do not rule out early KC, reinforcing the importance of biomechanical assessment in subclinical detection. Although they identified tomographic risk indicators in VAE, they also noted variability in biomechanical parameter performance across studies, underscoring the need for integrated indices.

Lu et al¹⁷ showed that combining spectral-domain OCT with air-puff tonometry in AI models significantly improved detection of eyes with subclinical KC. Their RF model achieved an AUC of 0.99 using both structural and biomechanical data. Notably, biomechanical parameters only (such as second applanation deflection amplitude, area, and length) outperformed tomographic features, further supporting the concept that biomechanical changes precede morphological alterations in early disease.

Beyond air-puff-based and OCT methods, other imaging modalities are increasingly contributing to cornea assessment. Motion-tracking Brillouin microscopy, which directly measures focal biomechanical changes, has shown superior accuracy compared to traditional tomographic and combined indices in distinguishing subclinical KC.⁵¹ This highlights the promise of high-

resolution biomechanical techniques for detecting early ectatic changes before they appear on topography. Additionally, studies using high-frequency ultrasound and ocular pulse elastography have demonstrated elevated corneal axial strains and cone-centered deformation in KC, particularly in advanced stages.^{12,13} These findings align with our results, reinforcing the clinical relevance of localized biomechanical abnormalities in disease progression.

Despite its strengths, this study has several limitations. First, the cross-sectional design may introduce selection bias; future studies should incorporate longitudinal follow-up to assess dynamic biomechanical changes over time,⁵² potentially using emerging AI models such as deep learning¹⁹ for temporal analysis.¹⁵ Second, although external validation was performed on a data set from a different racial and geographic population, the training and initial testing were limited to a single center, which may affect generalizability. Multicenter validation is essential to evaluate clinical performance and utility.⁷ Third, the study has not yet been integrated into clinical workflows; future work should focus on developing practical, user-friendly decision support tools for ophthalmologists. Finally, because of data limitations, we were unable to include post-LASIK E cases⁵³ in the VAE-NT group, restricting the applicability of our findings to refractive surgery risk assessment. Future studies should aim to include such cases to enhance model relevance in surgical screening contexts.

These findings hold important clinical potential. With further validation on larger, diverse data sets, the AI-derived index could serve as a decision support tool,

aiding corneal specialists in earlier and more accurate E diagnosis. Improved detection of subclinical/early E risk may lead to timely interventions, reducing visual impairment and enhancing patient quality of life.^{2,4} Additionally, given the Corvis instrument's more compact design and lower cost compared to the Pentacam, the AI-derived index is well-suited for integration into telehealth platforms, potentially expanding access to specialized care in underserved or remote areas.⁵⁴

Conclusion

This study highlights the superior performance of an AI-derived index based on spatiotemporal biomechanical features from Corvis ST videos in detecting very early E. By analyzing pixel-level displacement and correlation-based metrics, an index was derived that identifies subtle biomechanical changes that precede structural abnormalities and surpasses traditional morphology-based indices. This approach holds strong potential to enhance early diagnosis, guide clinical decision-making, and improve preoperative screening for refractive surgery. Future work should validate these findings in larger multicenter cohorts and explore emerging AI models such as deep learning and large language model integration to further advance early E detection.

Acknowledgments

This research was conducted by the cornea research network (CRN) initiative, an international effort to develop AI models to address corneal conditions.

Footnotes and Disclosures

Originally received: June 9, 2025.

Final revision: August 21, 2025.

Accepted: September 22, 2025.

Available online: September 30, 2025. Manuscript no. XOPS-D-25-00418.

¹ Department of Ophthalmology, Assiut University, Assiut, Egypt.

² Department of Ophthalmology, Eye Hospital, Ruhr University, Bochum, Germany.

³ Zentrum für Refraktive Chirurgie, Muenster, Germany.

⁴ Poostchi Ophthalmology Research Center, Shiraz University of Medical Sciences, Shiraz, Iran.

⁵ Biomedical Engineering Department, Al-Khwarizmi College of Engineering, University of Baghdad, Baghdad, Iraq.

⁶ Computers, Electronics and Automation Department, Stefan cel Mare University of Suceava, Suceava, Romania.

⁷ University of Western Brittany, Brest, France.

⁸ LaTIM UMR 1101 Inserm, Brest, France.

⁹ Center for Cyber Medicine Research, University of Tsukuba, Ibaraki, Japan.

¹⁰ Department of Ophthalmology, Jichi Medical University, Tochigi, Japan.

¹¹ Department of Ophthalmology and Visual Sciences, Universidade Federal de Sao Paulo Escola Paulista de Medicina, Sao Paulo, Sao Paulo, Brazil.

¹² Bascom Palmer Eye Institute, Department of Ophthalmology, University of Miami, Miami, Florida.

¹³ Department of Electrical and Computer Engineering, University of Miami, Miami, Florida.

Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The authors made the following disclosures:

S.T.: Consultant — Oculus.

H.T.: Founder — DeepEyeVision.

A.L. was supported by a grant of the Ministry of Research, Innovation and Digitization, under the Romania's National Recovery and Resilience Plan funded by the EU Next Generation EU program, project "Artificial intelligence-powered personalized health and genomics libraries for the analysis of long-term effects in COVID-19 patients (AI-PHGL-COVID)" number 760073/23.05.2023, code 285/30.11.2022, within Pillar III, Component C9, Investment 8; and by a grant of the Ministry of Research, Innovation and Digitization, CNCS, UEFISCDI, project number PN-III-P1-1.1-PD-2021-0273, within PNCIDI III.

HUMAN SUBJECTS: No human subjects were included in this study. Institutional review board offices of both centers including Ethik-Kommission der Ärztekammer Westfalen-Lippe and Salouti Cornea Research Center approved the retrospective analysis of anonymized patient data. The study adhered to the Declaration of Helsinki and its subsequent amendments. Written informed consent was obtained from participants prior to data use, and ethical approval was obtained from the institutional review

board offices of both centers. All patient data were fully deidentified prior to analysis in both countries to ensure confidentiality and compliance with data protection regulations.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Abdelmotaal, Yousefi

Data collection: Abdelmotaal, Taneri, Salouti, Nowroozzadeh, Al-Timemy, Lavric, Daho, Takahashi, Hazarbasanov, Yousefi

Analysis and interpretation: Abdelmotaal, Yousefi

Obtained funding: N/A

Overall responsibility: Abdelmotaal, Taneri, Salouti, Nowroozzadeh, Al-Timemy, Lavric, Daho, Takahashi, Hazarbasanov, Yousefi

Abbreviations and Acronyms:

AI = artificial intelligence; **AUC** = area under the curve; **AUSEP** = area under the separation curve; **BAD-D** = Belin/Ambrósio Enhanced Ectasia Display; **CBI** = Corvis Biomechanical Index; **CCT** = central corneal thickness; **CDVA** = corrected distance visual acuity; **D** = diopter;

DCR = dynamic corneal response; **E** = ectasia; **KC** = keratoconus; **Kmax** = maximum keratometry; **KNN** = K-nearest neighbors; **LVC** = laser vision correction; **MKC** = mild keratoconus; **ML** = machine learning; **N** = normal; **NT** = normal topography; **PCA** = principal component analysis; **RF** = random forest; **SD** = standard deviation; **TBI** = tomographic and biomechanical index; **TKC** = topographical keratoconus classification; **VAE** = very asymmetric ectasia; **VAE-NT** = very asymmetric ectasia with normal topography; **2D** = 2-dimensional.

Keywords:

Artificial intelligence, Corneal biomechanics, Early detection, Keratoconus.

Correspondence:

Hazem Abdelmotaal, MD, Department of Ophthalmology, Assiut University Hospital, Assiut University, Assiut 71516, Egypt. E-mail: hazem@aun.edu.eg; and Siamak Yousefi, PhD, McKnight Vision Research Center, 1638 NW 10th Avenue, Suite 824, Miami, FL 33136. E-mail: siamak.yousefi@miami.edu.

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